

What Lifestyle Factors Predict Diabetes Risk?

A Predictive Analysis

Group 7: Ayantu Bekelche, Adrien Rozario, Gavin Schneider, Grace Yu, Nimisokan Ojikutu, Xinyu Yuan

Table of contents

01

Introduction

02

Research Question

03

Background

04

Data Cleaning

05

Data Visualization

06

Cross Validation
Model Performance

Table of contents

07

**Fairness & Bias
Evaluation**

08

Test Set Evaluation

09

Metric Selection

10

**Recommendation
to Clinicians**

11

Conclusion

Background

- Diabetes is a common chronic disease in the United States impacting millions of Americans

Patients lose the ability to regulate levels of glucose (sugar) in the blood, leading to reduced quality of life and life expectancy

- Lifestyle factors may increase diabetes risk
 - BMI
 - Physical activity
 - Smoking
 - Alcohol consumption
- Many individuals may be unaware of their risk of diabetes and may benefit from accurate model predictions, allowing early detection and prevention



Research Question

Can we predict what lifestyle factors predict diabetes risk?

- Exploratory Data Analysis
 - Understanding linked factors can help with targeted intervention
- Predictive Analysis
 - Accurate early prediction can improve prevention strategies



Data Cleaning

Data:

- The Behavioral Risk Factor Surveillance System (BRFSS) is a health survey that is collected from over 400,000 Americans annually by the CDC
- We used the CDC Diabetes Health Indicators dataset from 2015
- 253,680 observations (patient records)
- 22 features (health + lifestyle variables)

Cleaning:

- Ensured no missing/inconsistent values
- Converted categorical variables from binary (0, 1) to labels (No, Yes)
- Converted ordinal variables to labeled factors (ex. ages)
- Standardized continuous variables with z-score normalization



Exploratory Data Analysis

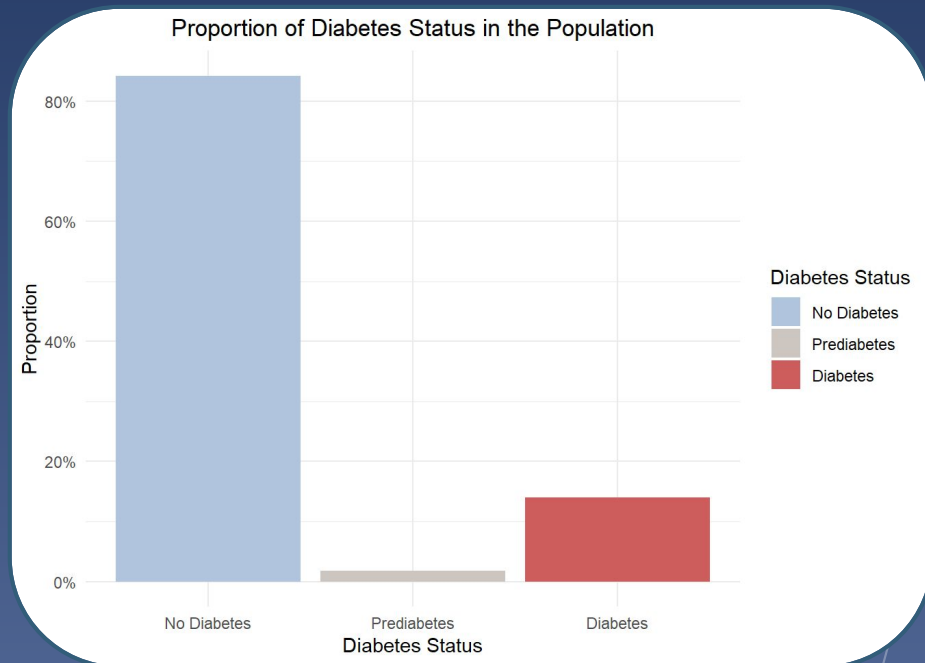
1. What proportion of individuals have diabetes?

Key Findings:

- Over 80% had no diabetes
- About 15% were diabetic
- Prediabetes was the smallest group
- Clear class imbalance in the dataset

Takeaway:

- Class imbalance may affect model performance
- Sensitivity and specificity are important metrics



Exploratory Data Analysis

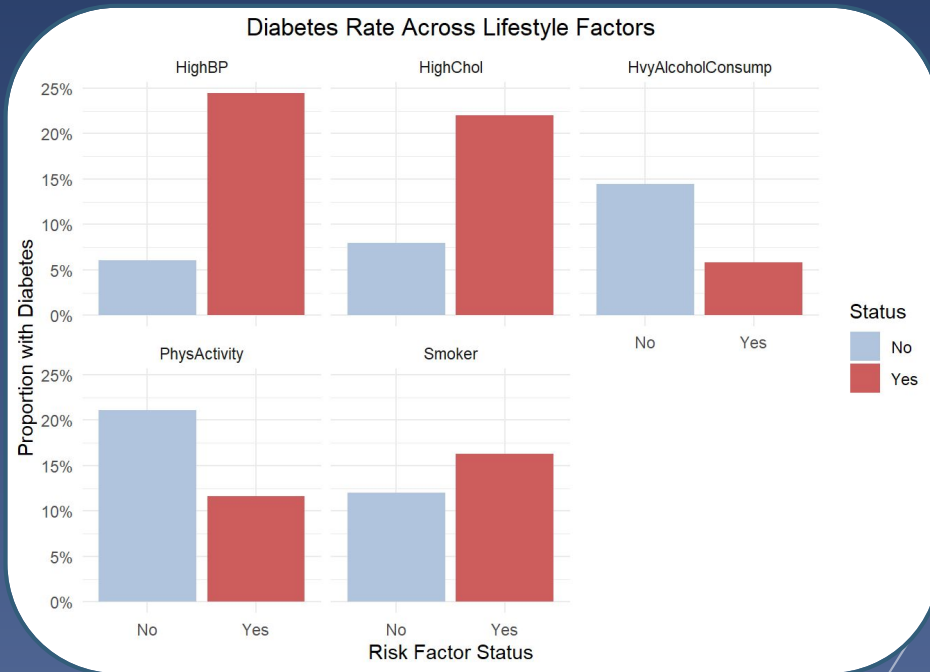
2. Which lifestyle factors appear most strongly associated with diabetes?

Key Findings:

- High blood pressure and high cholesterol showed the strongest association
- Physical activity was linked to lower diabetes rates
- Smoking and alcohol consumption had smaller effects

Takeaway:

- Cardiovascular health indicators were major predictors of diabetes risk
- Healthy lifestyle choices may reduce risk



Exploratory Data Analysis

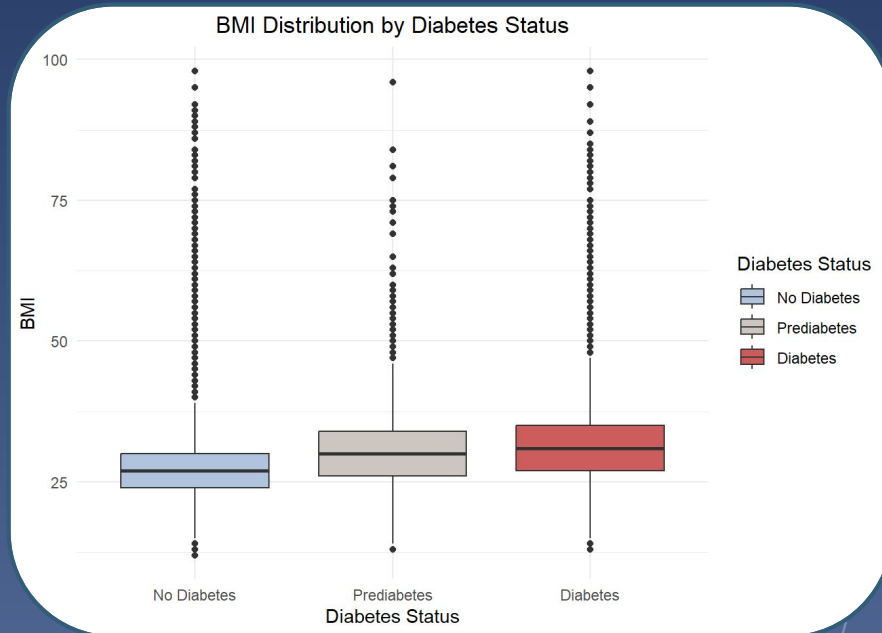
3. How does BMI differ between individuals with and without diabetes?

Key Findings:

- Diabetic individuals had higher BMI values
- Median BMI increased across diabetes groups
- BMI distributions overlapped across groups

Takeaway:

- Higher BMI was associated with slightly increased diabetes risk
- BMI alone does not fully predict diabetes outcomes



Exploratory Data Analysis

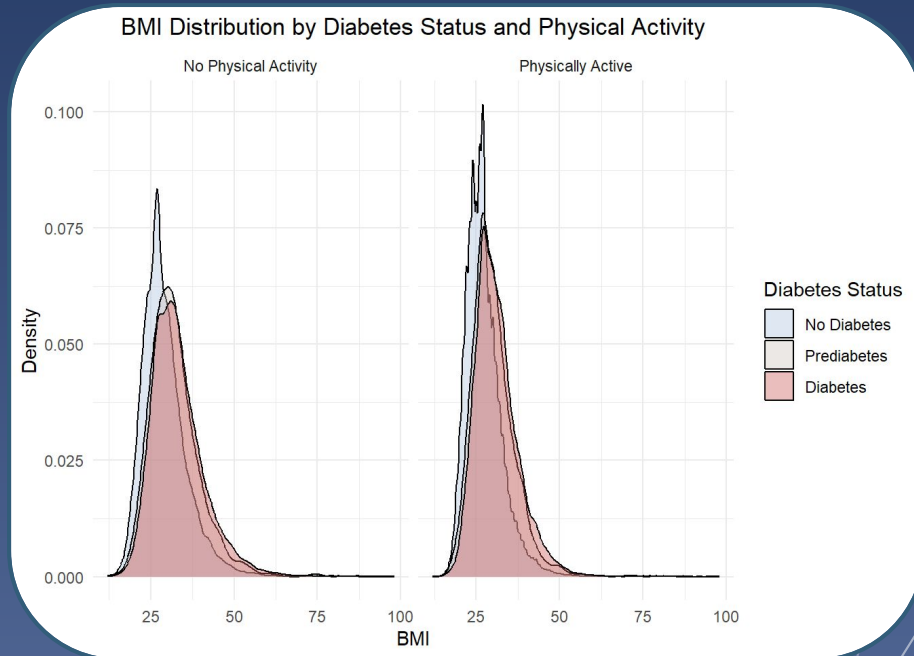
4. How do BMI and Physical activity interact to influence diabetes risk?

Key Findings:

- Diabetes groups consistently showed higher BMI
- Physically active individuals had slightly lower BMI
- Physical activity reduced risk patterns modestly

Takeaway:

- Exercise may reduce diabetes risk
- High BMI remained an important risk factor



Exploratory Data Analysis

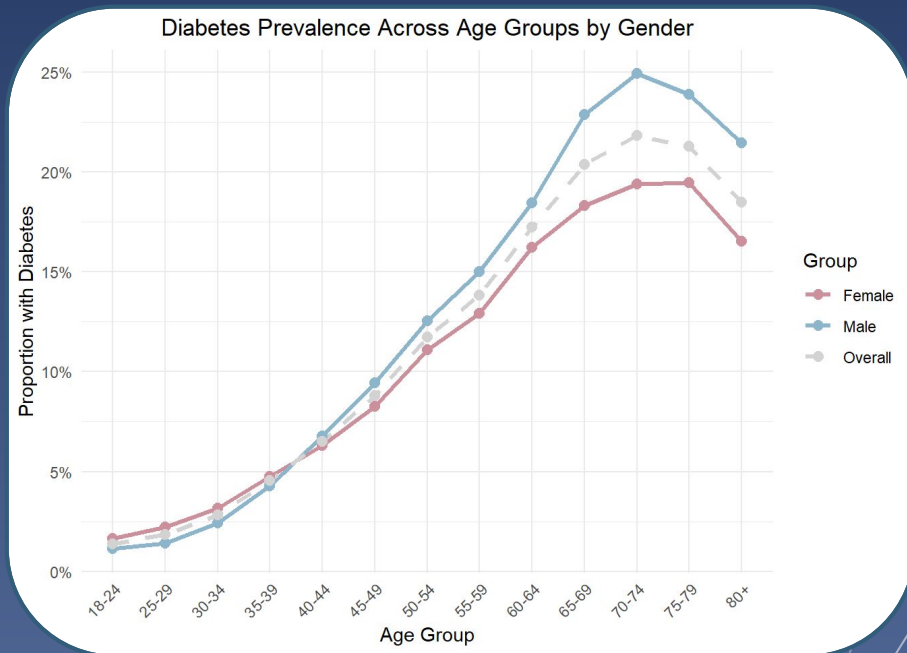
5. How does diabetes prevalence vary by age group and gender?

Key Findings:

- Diabetes prevalence increased with age
- Males had slightly higher diabetes risk than females across age groups

Takeaway:

- Age was one of the strongest predictors of diabetes risk
- Small demographic differences may still influence model fairness and prediction outcomes



Predictive Analysis

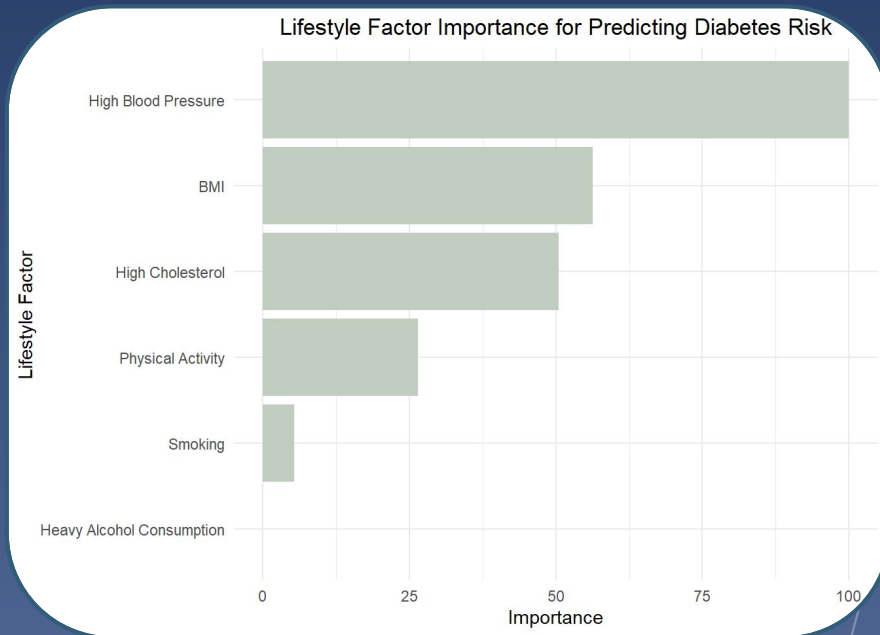
Ranking Lifestyle Factor Feature Importance in Predicting Diabetes Risk

Key Findings:

- Cardiovascular and weight-related indicators were most strongly linked to diabetes risk

Takeaway:

- Predictive modeling reinforced patterns observed during exploratory analysis
- Decision trees provided insights into which health factors most strongly influence diabetes classification

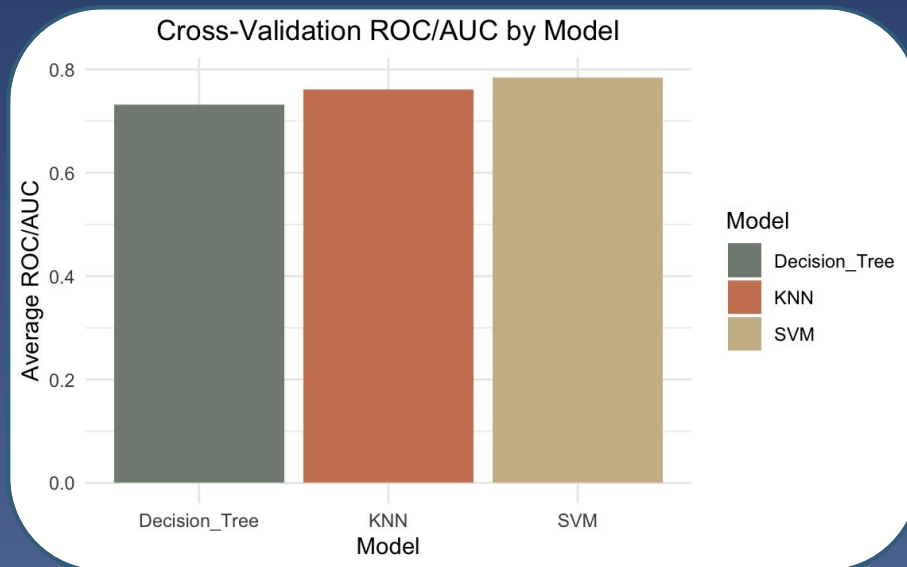


Cross-Validation Model Performance

Comparing Classification Model Accuracy with ROC/AUC Scores

In terms of classifying true positive diabetes diagnosis:

- Decision Tree had weakest performance (0.732)
- KNN had moderate performance (0.762)
- SVM had highest ROC/AUC score (0.784)
- SVM was most accurate at correctly classifying diabetes patients, so it is reliable for further fairness evaluation and testing



Fairness & Bias Evaluation

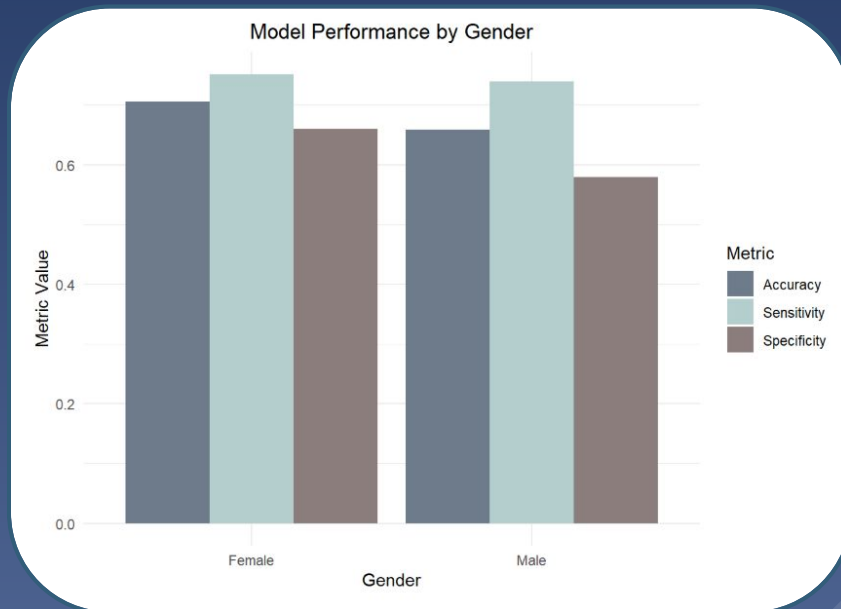
Model Performance by Gender

Key Findings:

- SVM Accuracy was slightly higher for females
- SVM Sensitivity was similar across genders
- SVM Specificity was lower for males
- The SVM model produced more false positives for male individuals

Takeaway:

- Small fairness differences existed across demographic groups
- Fairness evaluation is important in healthcare modeling



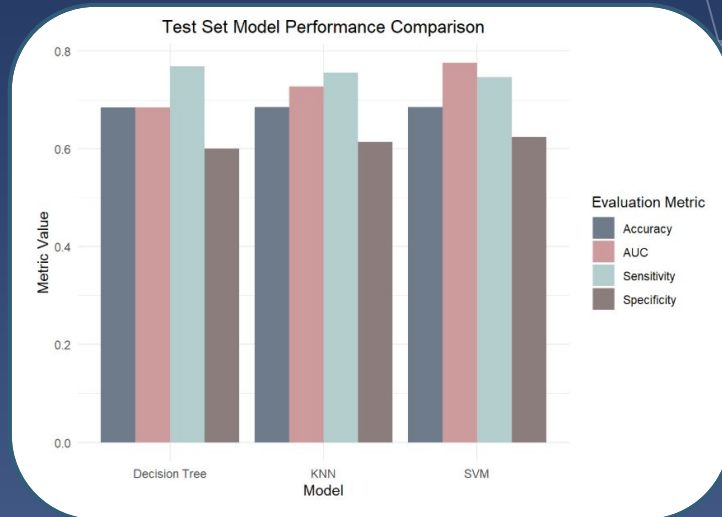
Test Set Model Evaluation

Key Findings:

- All models achieved similar accuracy (~68%)
- SVM achieved the highest AUC score (~0.77)
- KNN performed slightly better than Decision Tree
- Sensitivity was higher than specificity across models

Takeaway:

- SVM showed the strongest overall performance
- Models generalized well with minimal overfitting
- Lower specificity may increase false positives



Model	AUC	Accuracy	Sensitivity	Specificity
Decision Tree	0.684	0.684	0.768	0.600
SVM	0.776	0.685	0.746	0.624
KNN	0.727	0.685	0.756	0.614

Model Recommendation

Metric Selection

- Prioritize: Sensitivity (Recall)
- Identifies true diabetes cases
- Missing a diagnosis (false negative) → serious health risks
- False positives → less harmful (extra screening)

Recommendations to Clinicians

- Best Model: SVM
- Highest overall performance (AUC)
- Strong sensitivity → effective for screening

Considerations Before Use

- Moderate specificity → more false positives
- Slight gender bias in performance
- Needs:
 - More clinical variables
 - Fairness improvements

Conclusion

- Use SVM as a screening support tool, not a replacement for clinical judgment

Conclusion

Primary Findings

- 1 BMI, high blood pressure, and high cholesterol were strong predictors of diabetes risk
- 2 Physical activity was associated with lower diabetes prevalence
- 3 Age strongly influenced diabetes prevalence
- 4 SVM model produced the best predictive performance

Final Takeaway

- 1 We can prevent early diabetes risk by providing targeted intervention for groups with those lifestyle factors
- 2 Machine learning models like SVM can be used for diabetes risk prediction, but fairness and bias should remain important considerations



Thank You

Do you have any questions?

CREDITS: This presentation template was created by [Slidesgo](#), and includes icons by [Flaticon](#) and infographics & images by [Freepik](#)

Please keep this slide for attribution